

Solution

ELECTRICITY - 08

Class 10 - Science

1. Death may be caused if a heavy current flows through the body. Current flows not due to potential of the body but due to potential difference. When a person standing on earth touches a 220 V line, the potential difference of two points is 220 V [Since the potential of earth on which he stands is at 0 V potential] A bird however is at 11,000 V potential but no current flows through it.
2. When the fairy lights are connected in series the resistance offered will be greater and brightness of the bulbs will be affected. But in parallel connection all the bulbs will glow with same intensity and if any bulb gets fused the other bulbs will continue to glow.
3. Potential difference between two points in a current carrying conductor is 1 V if 1 joule of work is done to move a charge of 1 C from one point to the other.

$$1V = \frac{1J}{1C} \text{ or } 1V = 1JC^{-1}$$

Potential difference is measured by voltmeter. Voltmeter is always connected.

4. i. The potential at a point is defined as the amount of work done in bringing a unit positive charge from infinity to that point.
ii. The potential difference between two points is equal to the work done in moving a unit positive charge from one point to the other.

5. Here, the potential difference, $V = 2 + 2 + 2 = 6V$ and charge, $Q = 1 C$

We know that, Work done, $W = \frac{V}{Q}$

Substituting the value, we get

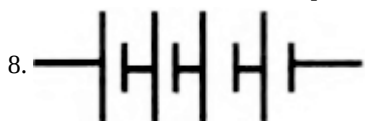
$$W = \frac{6}{1} = 6 J.$$

6. Energy to be given to 1 C of charge to pass through 1 V potential difference = 1 J.

Energy to be given to 1C of charge to pass through 6 V battery = 6 J

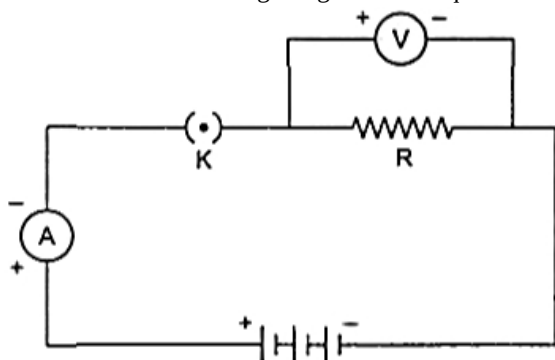
$$\text{or } \Omega = VQ = 6 \text{ volt} \times 1 \text{ coulomb} = 6 \frac{J}{C} \times 1C = 6 J$$

7. Zero, current flows due to potential difference and not due to potentials.

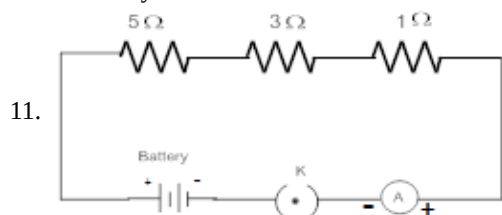


9. An electrical circuit is a continuous path comprising conducting wires and other electrical components between the terminals of a battery along which an electric current is set up. It is represented by drawing a circuit diagram.

The corrected circuit diagram given in the question is:



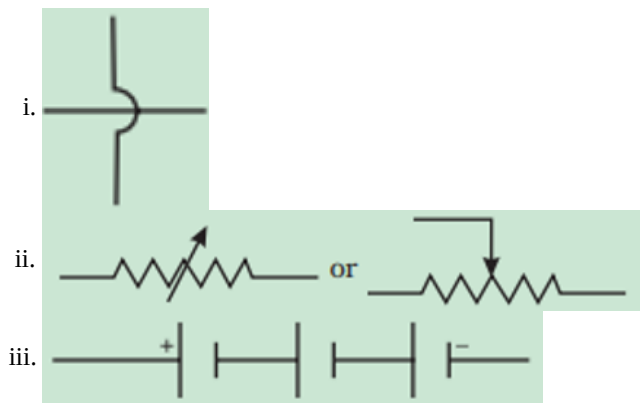
10. There is only a single path which connects from the electric source to the output devices. The electrical appliance damage can be prevented by connecting the fuse in series with the mains as well as the electrical appliance. To maintain the current level efficiently series of resistors can be used.



12. An electric circuit is said to be an open circuit when the switch is in 'off' mode (or key is unplugged) and no current flows in the circuit.

The circuit is said to be a closed circuit when the switch is in 'on' mode (or key is plugged) and a current flows in the circuit.

13. The symbols of the following components that are used in the circuit diagram are as follows:



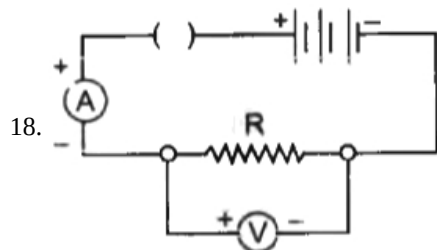
14. The potential difference across any two points in a circuit is measured by a voltmeter. A voltmeter must have a high resistance and it is connected in parallel to the two points across which the potential difference is to be measured.

15. Following precautions should be taken:

1. The ends of connecting wire should be neat and clean.
2. Never allow the current of flow in the resistance of wire for a long time.
3. Range of voltmeter should be greater than the applied voltage.
4. When not in use, supply of current should be switched off.
5. A low resistance rheostat must be used.

16. This is called the zero error of the scale of ammeter or voltmeter. If there is a zero error then this error is subtracted from the value that depicts when the circuit is closed otherwise accurate reading will not be recorded.

17. The current in a circuit is measured by an ammeter. Ammeter is always connected in series in a circuit and the positive marked terminal of the ammeter is connected towards the positive terminal of the source of current.

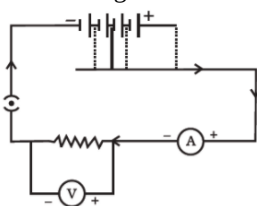


19. Here $I = 2.5 \text{ mA} = 2.5 \times 10^{-3} \text{ A}$; $V = 12 \text{ volt}$; $R = ?$

$$V = IR \text{ or } R = \frac{V}{I}$$

$$R = \frac{12}{2.5 \times 10^{-3}} = \frac{12}{25 \times 10^{-4}} = \frac{120000}{25} = 4800 \Omega$$

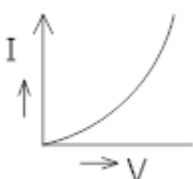
20. Correct diagram is as follows:



21. i. a. Ohmic material



b. Non-ohmic material



ii. Example of ohmic material - Copper, Nichrome etc.

Example of Non-ohmic material –Diode, Transistor made of semi-conductors like silicon, germanium etc.

22. Ohm's law is obeyed only when the temperature of conductor remains constant.

23. According to Ohm's law, physical conditions remaining constant, $V \propto I$ where V is the potential difference across the ends of conductor and I , the current flowing through it.

When potential difference is halved, the current will be halved since resistance remain constant.

$\therefore$ For same resistance, current is half of the former.

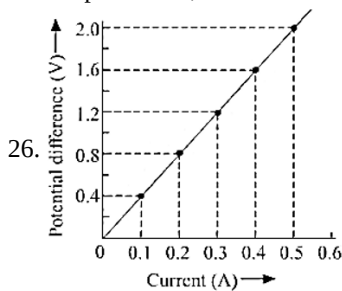
24. $I = 30 \text{ mA} = 30 \times 10^{-3} \text{ A}$, $R = 50 \Omega$; $V = ?$

$$V = IR = 30 \times 10^{-3} \times 50 = 1,500 \times 10^{-3} \text{ or } V = 1.5 \text{ volt}$$

25. According to Ohm's law, the current (I) flowing through a conductor is directly proportional to the potential difference (V) across its ends, provided its physical conditions remain the same.

Ohm's law does not hold good under all conditions as it is not a fundamental law of nature like Newton's laws.

It is obeyed by metallic conductors only when physical conditions like temperature etc. are kept unchanged. It is not obeyed by a lamp filament, Junction diode, thermistor, etc.



The graph between V and I is a straight line.

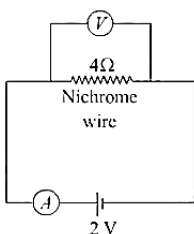
So we can conclude that current flowing through the wire is directly proportional to the potential difference V across it.

The resistance of the wire can be calculated as

$$R = \frac{V}{I} = \frac{0.4}{0.1} = 4\Omega$$

We can also conclude that the nichrome wire has a constant value of the resistance as 4Ω .

Nichrome wire follows ohm's law thus it behaves like an ohmic conductor. The circuit diagram corresponding to the above graph is as shown below.

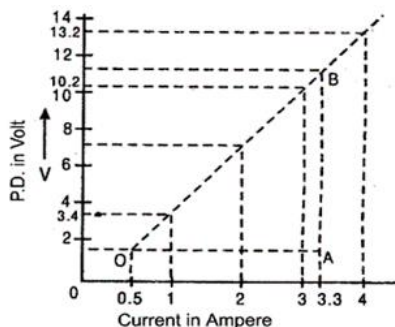


27. Graph between I and V is as shown in figure. Since graph is almost a straight line, the slope of line between any two points give the resistance. Slope of say OB is:

$$R = \frac{AB}{OA} = \frac{BC-AC}{DA-OD}$$

$$\frac{AB}{OA} = \frac{12-1.6}{3.3-0.5} = \frac{10.4}{2.8} = \frac{104}{28}$$

$$R = 3.7\Omega$$



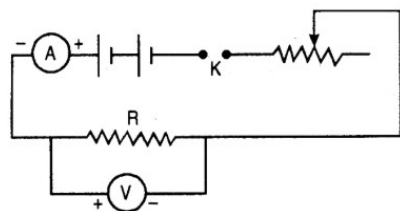
28. Since the straight line A is steeper than B, so the straight line A represents a greater resistance. The equivalent resistance in a series combination is greater than in parallel combination, hence A represents series combination.

29. Ohm's law states that the current passing through a conductor is directly proportional to the potential difference across its ends provided the temperature and other physical conditions remains unchanged.

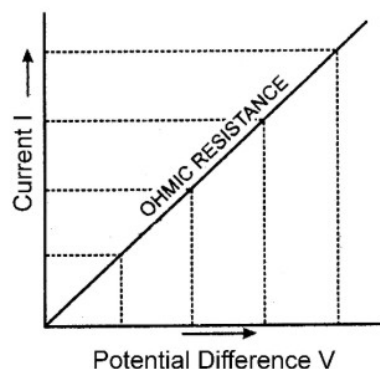
$$\text{or } I \propto V \text{ or } V \propto I \text{ or } V = RI$$

where R is a constant called resistance. Resistance is the property of a body to oppose the flow of current. R the resistance depends upon the nature of the conductor, its temperature and its dimensions (length, area)

$$R = \frac{V}{I} \text{ or } I = \frac{V}{R} \text{ i.e. } I \propto V \text{ and } I \propto \frac{1}{R}$$



Experimental Verification of Ohm's Law: To verify Ohm's law, take a resistor R, connect voltmeter across it. Connect an ammeter, battery, key and rheostat to it as shown in Fig. Put in the key K. Read the value of potential difference across resistor R with the help of voltmeter and the current flowing through resistor with the help of ammeter. Note the readings. Vary the current in the circuit by sliding rheostat and go on noting reading in voltmeter and ammeter. Plot a graph between V and I on graph paper. It will come out to be straight line as shown in fig.



30. $I=10 \text{ A}$, $V=220 \text{ V}$

$$R = \frac{V}{I}$$

$$= \frac{220}{10}$$

$$= 22 \Omega$$

31.	Ohmic Resistor	Non-Ohmic Resistor
	Ohmic resistors obey Ohm's law.	Non-Ohmic resistors do not obey Ohm's law.
	The graph for potential difference (V) versus current (I) is a straight line.	The graph for potential difference (V) versus current (I) is not a straight line.
	The slope of V-I graph is constant at all values of V or I at a given temperature.	The slope of V-I graph is different at different values of V or I even at a given temperature.

32. If temperature and other physical conditions such as pressure, mechanical strain, etc. Remain the same, the current (I) flowing through a conductor is directly proportional to be potential difference (V) across the conductor, i.e.,

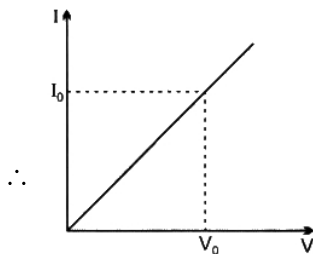
$$I \propto V \text{ or } V = IR$$

Where R is a constant called resistance.

If a graph is plotted between the current (I) flowing through the condutor and the applied potential difference (V) between its ends, it will be straight line.

33. The straight line in the graph signify that potential difference and current are directly proportional to each other. The method of determining resistance of resistor using the graph is by Ohm's law, $V = IR$ and by calculating the slope from the points mentioned

on the graph.



$$R = \frac{1}{\text{Slope of V-I graph}}$$

34. The figure (a) obeys Ohm's law because the V-I graph is a straight line and its slope (or resistance) is constant.

35. The figure (b) and (d) do not obey Ohm's law because the V-I graphs are not straight lines.

36. $R_S = R_3 + R_4 = 10 + 10 = 20 \Omega$

$$\frac{1}{R_P} = \frac{1}{R_2} + \frac{1}{R_S}$$

$$= \frac{1}{20} + \frac{1}{20} = \frac{1}{10} \Omega$$

$$R_P = 10 \Omega$$

$$\text{Total equivalent resistance} = R = R_1 + R_P + R_5$$

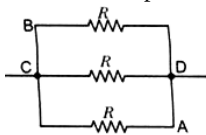
$$= R = 20 + 10 + 10 = 40 \Omega$$

37. i. Parallel connection is better in domestic circuit because each device gets same potential, it keeps on working even, if other devices stop working, parallel connection is also helpful when each device has different resistances .

ii. Electrician has made series connection of all the lamps in electric circuit of house because of which, if one lamp gets fused, all the other lamps stop working.

This is due to the fact that when devices are connected in series, then if one device fails, the circuit gets broken and all the devices in that circuit stop working.

38. The point C is connected to B and the point D is connected to A. Therefore, three identical resistors, each having resistance R, are connected in parallel and the equivalent circuit diagram is shown in the figure. If the equivalent resistance is R then

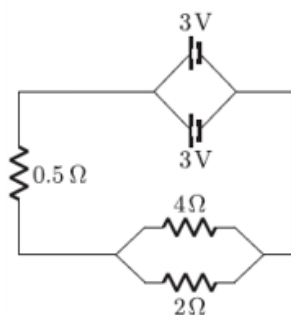


$$\frac{1}{R'} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

$$= \frac{3}{R}$$

$$\Rightarrow R' = \frac{R}{3}$$

39. The circuit diagram is as follows:



To calculate the current

Resistor 4Ω and 2Ω are connected in parallel. So, their equivalent resistance is given by

$$R_p = \frac{4 \times 2}{4 + 2} = \frac{8}{6} = \frac{4}{3} \Omega = 1.33 \Omega$$

Total resistance of circuit,

$$R = R_p + 0.5 \Omega = (1.33 + 0.5) \Omega = 1.83 \Omega$$

Current in the circuit,

$$I = \frac{3V}{1.83\Omega} = 1.64A$$

Potential difference across 0.5Ω resistor is

$$V' = 1.64 \times 0.5 = 0.82 V$$

The potential difference across 4Ω resistor is

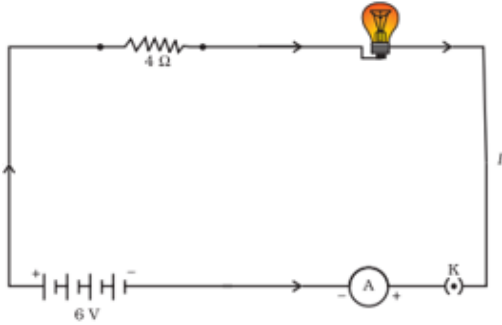
$$V'' = V - V' = 3 - 0.82 = 2.18 \text{ V}$$

Thus, current flowing through 4Ω resistor is

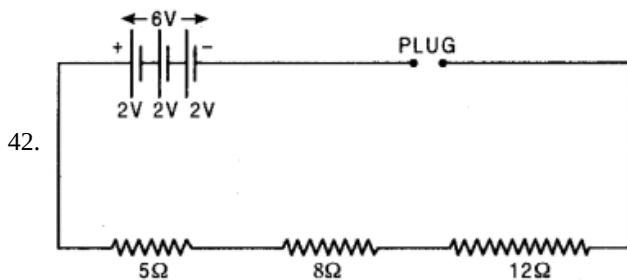
$$I_1 = \frac{2.18\text{V}}{4\Omega} = 0.55\text{A}$$

40. a. No, the bulb will not glow with same brightness in both the circuits as voltage gets divided in series combination, So bulbs in series combination will glow with less brightness. Voltage does not get divided in parallel combination, So bulbs in parallel combination will glow with more brightness.
- b. In case of series combination, fault in even a single component will break the circuit. So, when one of the bulbs gets fused; another bulb would not continue to glow.

41. Figure: An electric lamp connected in series with a resistor of 4Ω to a 6 V battery.



- a. The resistance of electric lamp, $R_1 = 20\Omega$,
The resistance of the conductor connected in series, $R_2 = 4\Omega$.
Then the total resistance in the circuit $R = R_1 + R_2 = 20\Omega + 4\Omega = 24\Omega$.
The total potential difference across the two terminals of the battery $V = 6\text{ V}$.
- b. Now by Ohm's law, the current through the circuit is given by $I = V/R_s = 6\text{ V}/24\Omega = 0.25\text{ A}$.



43. The resistance of an ammeter is very low. If an ammeter is connected in parallel, the resultant resistance of the circuit decreases and excessive current passes through the instrument. Hence, the ammeter is likely to be burnt out.
44. In series combination $R_s = nR$
For parallel $R_p = \frac{R}{n}$
 $\frac{R_s}{R_p} = \frac{nR}{\frac{R}{n}} = n^2$
 $\therefore$ The required ratio is $n^2:1$

45. It is given that potential difference (V) = 220 V .

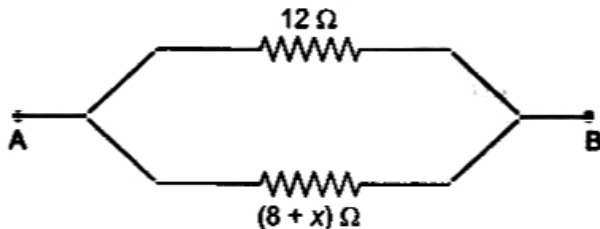
Resistance of coil A (R_A) = Resistance of coil B (R_B) = 24Ω

- i. When either coil A or B is used separately, the current (I) = $\frac{V}{R} = \frac{220\text{V}}{24\Omega} = 9.2\text{ A}$.
- ii. When two coils are used in series, total resistance, $R_s = R_A + R_B = 24 + 24 = 48\Omega$
Current flowing (I) = $\frac{V}{R_s} = \frac{220\text{V}}{48\Omega} = 4.6\text{ A}$
- iii. When the two coils are joined in parallel, total resistance (R_p) = $\frac{1}{24} + \frac{1}{24} = \frac{2}{24}$
 $R_p = 12\Omega$
Current (I) = $V/R_p = \frac{220\text{V}}{12\Omega} = 18.3\text{ A}$

46. We know $R \propto l$. In series combination of resistors, the effective length of cross-section of the conductor increase, so the resistance increases. In series combination the total resistance is equal to the sum of the individual resistances. Hence, the effective resistance of the circuit increases.

$$R_S = R_1 + R_2 + \dots + R_n$$

47. The network of resistors can be simplified as



($\because 5\Omega$ and 7Ω are in series and also 8Ω and $x\Omega$ are in series). Here, equivalent resistance (R) = 6Ω , $R_1 = 12\Omega$, $R_2 = (8 + x)\Omega$.

The resistors R_1 and R_2 are connected in parallel.

$$\therefore \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\text{or } \frac{1}{6} = \frac{1}{12} + \frac{1}{8+x}$$

$$\text{or } \frac{1}{6} - \frac{1}{12} = \frac{1}{8+x}$$

$$\text{or } \frac{2-1}{12} = \frac{1}{8+x}$$

$$\text{or } \frac{1}{12} = \frac{1}{8+x}$$

$$\text{or } 8 + x = 12$$

$$\therefore x = 12 - 8 = 4\Omega$$

- 48.
- Take three resistors R_1 , R_2 , and R_3 and connect them in parallel to make a circuit.
 - Use voltmeter to take reading of potential difference R_1 , R_2 and R_3 .
 - Now, remove the resistor one by one and take the reading of potential difference of remaining resistors'/resistor.

Observation: Voltmeter reading was same in each case. This shows that the same potential difference exists across three resistors.

49. Given, resistance, $R = 2\Omega$

Maximum power, $P_{\max} = 18W$

Maximum current, $I_{\max} = ?$

As we know, $P = I^2 R$

$$\Rightarrow I = \sqrt{\frac{P}{R}}$$

$$= \sqrt{\frac{18}{2}} = 3A = I_{\max}$$

Maximum current that can flow through 2Ω resistor is 3 A. This current divides along B and C because they are in parallel combination. Voltage across B and C remain same and hence $\propto \frac{1}{R}$. Since, B and C have same resistance same current flows through them.

i.e. $\frac{3}{2} = 1.5$ A flows through B and C

50. Power (P_1) = $\frac{V^2}{R}$ (consumption by R)

Power (P_2) = $\frac{V^2}{2R}$ (consumption by 2R)

$$\text{Ratio } \frac{P_1}{P_2} = \frac{V^2}{R} \times \frac{2R}{V^2} = 2 : 1$$

51. We know that $R \propto \frac{1}{A}$

In parallel combination of resistors, the effective area of cross-section of the conductor increases, so the resistance decreases. It is also to be noted that, when resistances are connected in parallel, the current in the circuit gets parallel branches to go from one point to another across which the potential difference is applied. The equivalent resistance is the reciprocal of individual resistances. So, the equivalent resistance is less when resistors are connected in parallel.

52. Equivalent resistance of 6Ω and 4Ω in series,

$$R_1 = (4 + 6)\Omega = 10\Omega$$

Equivalent resistance of 8Ω and $x\Omega$ in series,

$$R_2 = (8 + x)\Omega$$

Now, R_1 and R_2 are in parallel. Therefore, the equivalent resistance of R_1 and R_2 connected in parallel can be calculated as

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\text{or } \frac{1}{5} = \frac{1}{10} + \frac{1}{8+x} \quad [\because R = 5\Omega]$$

$$\text{or } \frac{1}{5} - \frac{1}{10} = \frac{1}{8+x}$$

$$\text{or } \frac{2-1}{10} = \frac{1}{8+x}$$

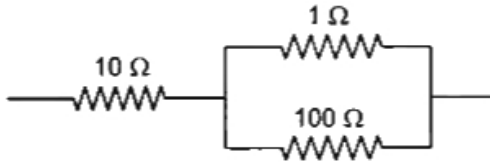
$$\text{or } \frac{1}{10} = \frac{1}{8+x}$$

$$\text{or } 8 + x = 10$$

$$\Rightarrow x = 10 - 8 = 2\Omega$$

$\therefore$ Value of $x = 2\Omega$

53. The resistors have to be combined as shown in the diagram..



The equivalent resistance of 1Ω and 100Ω connected in parallel is,

$$\frac{1}{R_1} = \frac{1}{1} + \frac{1}{100}$$

$$= \frac{100+1}{100}$$

$$= \frac{1}{100}$$

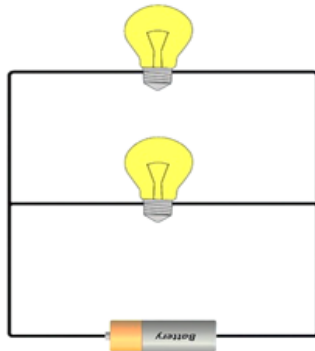
$$R_1 = \frac{100}{101}\Omega = 0.99\Omega$$

or Now, 10Ω and R_1 are in series. Therefore, equivalent resistance R is

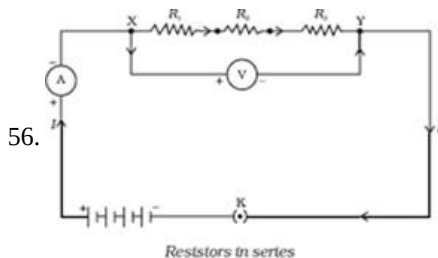
$R = (10 + 0.99) 10.99\Omega$ The above value of resistance is greater than 10.99Ω but less than 11Ω

54. A stronger current flows when the two resistors are connected in parallel, because in parallel combination the equivalent resistance of the two resistors will be less than that in series combination.

55. **Parallel Connection:** The electrical appliances should always be connected in parallel combination because if there is any fault/damage in any of the appliances then in series combination the whole circuit will get open but in parallel combination, only that part containing the damaged appliance is open which will have no effect on the working of other appliances in the circuit.



- Series Connection:** We do not use the series combination for connecting electrical appliances in the household circuit as whenever there will be a damage/breakage in the circuit of any one appliance of the household then, due to the series connection all other connections will also break.



56.

Take three resistances R_1 , R_2 and R_3 and Connect them in series combination to complete a circuit; as shown in figure. First of all, connect the ammeter between battery and R_1 After that, connect the ammeter between R_1 and R_2 , then connect the ammeter between R_2 and R_3 and take the readings one by one in all 3 cases.

Observation: We observed that ammeter shows same reading in all 3 cases. This shows that same current flows through every part of the circuit containing three resistances in series.

57. Because series arrangements have many disadvantages :-

- In series circuit if one electrical appliance stop working due to some defect , then all appliances stop working.
- For all the devices that are connected in series have only one switch due to which they cannot be turned off or on separately.

iii. In series circuit all electrical appliances do not get the same voltage (220V) because the voltage is shared by all the appliances.

iv. In series arrangement, the overall resistance increases to much due to which current from power supply is low.

58. i. $R_{AC} = R$

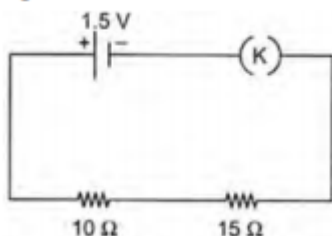
ii. The potential of B and D when the voltage source is applied across A and C remains the same.

iii. The potential of B and D when the voltage source is applied across A and B is different.

59. Advantages of parallel wiring in the domestic circuit are:

1. Voltage supply remains the same for all appliances connected in parallel.
2. If one appliance is not in order, it is not going to affect the working of other appliances.
3. If one appliance is turned on then other appliances can remain off saving lot of power.

60. i. The schematic circuit diagram is shown below:



ii. $1\Omega, 3\Omega, 2\Omega$ resistances are connected in series.

$$R = R_1 + R_2 + R_3$$

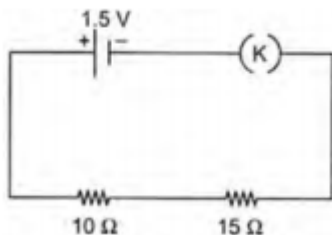
$$= 1 + 3 + 2 = 6\Omega$$

$$I = \frac{V}{R} = \frac{3}{6} = \frac{1}{2}A$$

Current in each resistance is same, i.e. $\frac{1}{2}A$.

Voltage across B & C = $\frac{1}{2} \times 3 = \frac{3}{2}$ Volt.

$$V_{BC} = \frac{3}{2} \text{ Volt}$$



61. a. Resistance is maximum when they are connected in series.

Therefore, highest resistance = $4 + 8 + 12 + 24 = 48\Omega$

b. Resistance is lowest when they are connected in parallel.

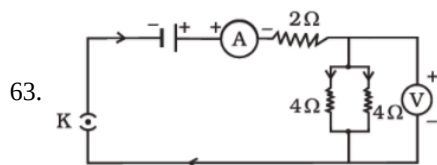
$$\frac{1}{R_p} = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{24} = \frac{6+3+2+1}{24} = \frac{12}{24} = \frac{1}{2}$$

$R_p = 2\Omega$ which is the lowest total resistance.

62. i. To get the smallest resistance, all the 5Ω resistors must be connected in parallel.

$$\text{Smallest resistance} = \frac{5}{50} = 0.1\Omega$$

ii. To get the largest resistance, the 5Ω resistors must be connected in series. Largest resistance = $5 \times 50 = 250\Omega$.



Total resistance for parallel combination of 4Ω resistor can be calculated as follow:

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

or, $R = 2\Omega$

Thus, resistance of parallel combination is equal to resistance of resistors in series. So, potential difference across 2Ω resistance will be same as potential difference across the other two resistors which are connected in parallel.

64. i. Let the resistance of the lamp = R_1

Resistance of conductor = $R_2 = 5\Omega$

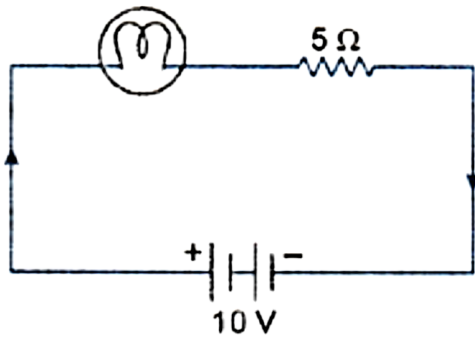
Total resistance in series, $R_S = R_1 + R_2 = R_1 = 5$

Current $I = 1$ A, Voltage, $V = 10$ V

Using Ohm's Law, $V = IR$

$$10 = 1 (R_1 + 5)$$

$$\Rightarrow R_1 = 5 \Omega$$



ii. Now, a resistance of 10Ω is connected in parallel with the series combination. Therefore, the total resistance of the circuit is given by

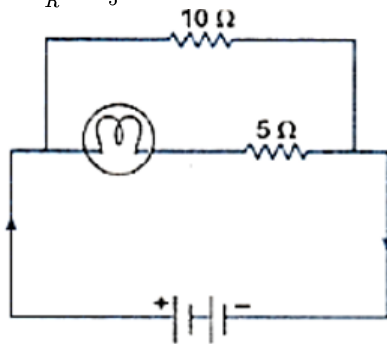
$$\frac{1}{R_P} = \frac{1}{R_1 + 5} + \frac{1}{10}$$

$$\Rightarrow \frac{1}{R_P} = \frac{1}{10} + \frac{1}{10}$$

$$\Rightarrow R_P = 5 \Omega$$

Hence, current flowing in the circuit,

$$I = \frac{V}{R} = \frac{10}{5} = 2 \text{ A}$$



Thus, 1 A current will flow through 10Ω resistor and 1 A will flow through the lamp and conductor of 5Ω resistance. Hence, there will be no change in current flowing through 5Ω conductor. Also, there will be no change in potential difference across the lamp.

65. Combined resistance of 100Ω , 50Ω and 500Ω in parallel

i.e. R_p is given by

$$\frac{1}{R_p} = \frac{1}{100} + \frac{1}{50} + \frac{1}{500} = \frac{5 + 10 + 1}{500} = \frac{16}{500}$$

$$R_p = \frac{125}{4} = 31.25 \Omega$$

Resistance of electric iron = 31.25Ω

$$\text{Current through electric iron} = \frac{V}{R} = \frac{220}{31.25} = 7.04 \text{ A}$$

